



School of Engineering
Diploma in AI & Data Engineering

EGDT305

Big Data Processing Platforms &
Applications

Big Data Analytics & Comparison
Report

AY2026S1

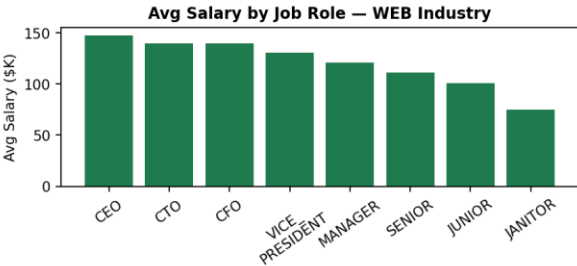
1. Perform data cleaning on PySpark

Summary: 1,000,000 to 991,997 rows (0.8% removed): 104 duplicate jobIds, 4 fabricated 'SCAMMER' rows, nulls dropped/imputed (median/'UNKNOWN'), distanceFromCBD outliers to 500,000km removed via IQR, salaryInThousands outliers removed via IQR (bounds \$11K-\$219K; one row read \$10,000,000). Full justification and before/after counts are in the submitted notebook, Section 3.

2. Exploratory Data Analysis.

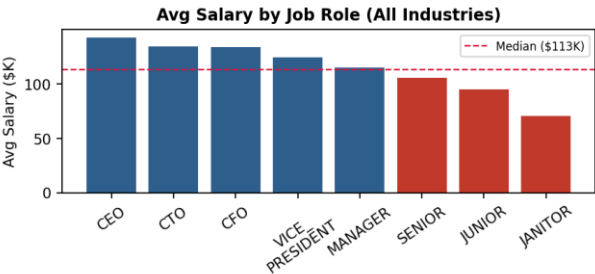
a. What is the highest paying job role for the web industry?

Answer: CEO is the highest-paying role in the WEB industry, averaging \$148K. Reason: executive roles carry the largest pay premium in every industry, and WEB is itself the 3rd highest-paying industry overall, so its top role clears \$140K.



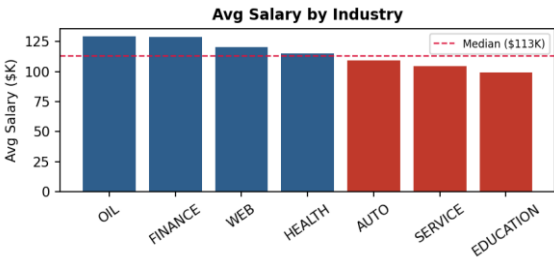
b. Rank the top 10 jobs roles with the highest salary for all the industry?

Answer: Only 9 distinct job roles exist in the cleaned dataset (no 10th): 1. CEO \$143K, 2. CTO \$134K, 3. CFO \$134K, 4. VICE_PRESIDENT \$125K, 5. MANAGER \$115K, 6. SENIOR \$105K, 7. JUNIOR \$95K, 8. PRESIDENT \$81K (n=1, statistically negligible), 9. JANITOR \$71K. Reason: the ranking mirrors organisational hierarchy almost exactly.



c. Which of the industries has the highest salary?

Answer: OIL has the highest average salary at \$129K, marginally ahead of FINANCE (\$129K). Reason: both are capital-intensive industries with high revenue per employee, which supports higher compensation.



d. Which job has the lowest pay?

Answer: JANITOR has the lowest average salary at \$71K. Reason: it is a non-specialised role with lower skill requirements and a larger available labour pool.

e. Which industries have the lowest pay?

Answer: GOVERNMENT is technically lowest (\$81K), but that is a single record in the cleaned dataset and is not a reliable finding. Among industries with adequate sample size, EDUCATION (\$99K) and SERVICE (\$104K) are the genuinely lowest-paying sectors.

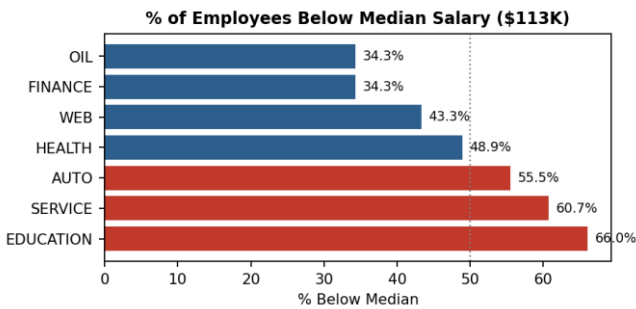
Reason: both are service driven, often publicly funded sectors rather than revenue/capital-intensive ones, which structurally caps salary budgets.

f. Median salary per year

Answer: The computed median salary is \$113,000/year.

g. Which industry has the highest percentage of people who are below the median salary?

Answer: Technically GOVERNMENT (100%), but this is a single-record category and not a meaningful finding. Among industries with adequate sample size, EDUCATION has the highest share below the \$113K median at 66.0%, followed by SERVICE (60.7%) and AUTO (55.5%). Reason: EDUCATION and SERVICE are lower revenue-intensity, often publicly-funded sectors, which structurally caps pay below the market median.



h. What are the job roles that are below the median salary?

Answer: SENIOR (\$105K), JUNIOR (\$95K), and JANITOR (\$71K) fall below the \$113K median. PRESIDENT (\$81K) is also technically below it but is a single-record role and not representative.

Reason: these sit in the lower rungs of the seniority ladder, where lower responsibility and a larger available labour supply keep pay below the market median (red bars in the job-role chart above).

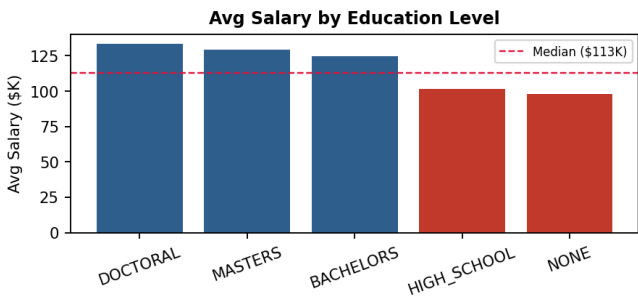
i. Determine if there is a relationship between years of experience and salary.

Answer: Yes, a positive but moderate relationship: Pearson correlation $r = 0.36$. Average salary rises steadily from \$92K at 0 years to \$137K at 24 years, but the wide spread of salaries within each experience level shows job role and industry are stronger predictors than tenure alone. (As a side comparison, distanceFromCBD correlates at $r = -0.29$ -- salary tends to be higher closer to the CBD.)



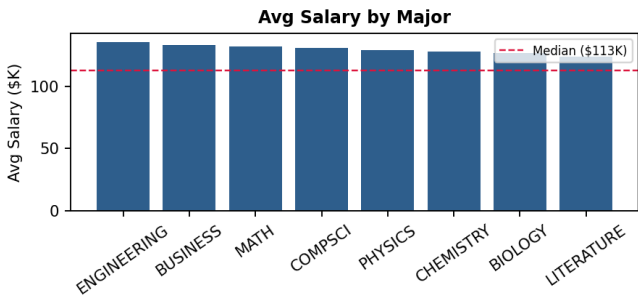
j. Is there a relationship between education and salary?

Answer: Yes: DOCTORAL \$134K > MASTERS \$129K > BACHELORS \$125K > HIGH_SCHOOL \$102K > NONE \$98K -- a clear, monotonic ordering. However the premium is modest (~\$9K from Bachelor's to Doctoral), so education is a contributing factor but not the dominant salary driver.



k. Did the major they studied affect the salary?

Answer: Yes, but the effect is small. ENGINEERING (\$136K) and BUSINESS (\$134K) pay the most, LITERATURE (\$124K) the least -- a spread of only ~\$12K, versus a ~\$72K spread across job roles. Major matters far less than role or industry. Overall conclusion across (i)-(k): job role and industry are the dominant salary predictors; education, experience, and major are secondary contributing factors.



3. ML Model Development using PySpark

Feature Engineering & Training Setup

- l. Categorical features (jobRole, education, major, industry) encoded via StringIndexer -> OneHotEncoder; numeric features (yearsExperience, distanceFromCBD) used as-is; all assembled into a 35-dimension feature vector.
- m. 80/20 train-test split (seed=42): 793,858 training rows, 198,139 test rows.
- n. Three regressors trained and compared: Linear Regression (regParam=0.1, elasticNetParam=0.8), Random Forest (100 trees, maxDepth=10), and Gradient Boosted Trees (100 iterations, maxDepth=5), each evaluated on the held-out test set with RMSE, R^2 and MAE.

Model Comparison (held-out test set)

Model	RMSE	R^2	MAE
Linear Regression	19.12	0.7385	15.53
Random Forest	20.52	0.6990	16.50
GBT Regressor (selected)	18.54	0.7543	15.14

Model selected: GBT Regressor -- lowest RMSE (\$18.54K), highest R^2 (0.754), lowest MAE (\$15.14K).

Reason: salary depends on interactions between categorical features (a CEO in OIL != a CEO in WEB premium). GBT's sequential error-correction -- each tree targets the previous ensemble's residuals -- captures these interactions, which a single linear equation cannot.

Why the other two were not selected: Random Forest scored below Linear Regression on every metric: with only 6 raw features, its random feature subsampling per split is too aggressive to consistently learn jobRole/industry/education interactions. Linear Regression was competitive ($R^2 = 0.74$, salary is mostly linear in experience) but can't model non-linear salary banding between roles, and has the highest MAE of the three.

Hyperparameter Tuning: GBT tuned via 3-fold cross-validation over maxDepth [4,5] x maxIter [50,100] (12 fits). Best: maxDepth=5, maxIter=100 -- RMSE=18.54, $R^2=0.754$, MAE=15.14, confirming the original hyperparameters were already near-optimal, now with cross-validated evidence.

4. Recommend strategies for how a person can chart their career

The profile of the person is:

- Years of experience in the following:
 - Software, AI & Data: 6
 - Backend Services: 14
- Interest: Board games, fixing things, Trading card games
- Yearly salary: \$100,000
- Married with three cats, one toddler, six chickens and one whole room of trading card games and blind boxes.
- Owns a house with a monthly mortgage of \$2,50
- Monthly household expenditure: \$6,000

Financial Baseline

Monthly mortgage (\$2,500) + monthly household expenditure (\$6,000) = \$8,500/month = \$102,000/year in committed expenses, against a current salary of \$100,000. That is already a \$2,000/year shortfall before accounting for the toddler's future costs, so salary growth is not optional, but it is financially necessary.

1. Which industry should this person move into?

Answer: FINANCE (fintech). It is the 2nd highest-paying industry in this dataset (\$129K average, vs. \$100K current), and it is the most natural landing spot for a 20-year Backend Services + Software/AI/Data profile.

2. Why that industry?

1. High, sustained demand for real-time data pipelines and ML models in fraud detection, algorithmic trading and risk analytics, where backend + data/AI skills intersect.
2. 14 years of backend services experience transfers directly to building the scalable, reliable systems banks and fintechs run on.
3. Singapore's financial sector is short on experienced engineers who also understand regulatory/data-integrity requirements (MAS, Basel) a 20-year veteran is a rare, valuable combination rather than a generic hire.
4. Singapore's ICT sector has an IMDA-identified gap of ~19,000 professionals; software engineers, data scientists and finance professionals are the most in-demand roles into 2026 (Jobs in Demand in Singapore 2026, finalroundai.com).

3. What skillsets are needed?

1. Cloud architecture (AWS/GCP): Singapore cloud engineers earn a median ~S\$110K, mid-level S\$105-145K, seniors S\$150-200K (skillup.sg, 2026) - certification is a direct lever on pay.
2. MLOps / data-engineering tooling (Spark, Kafka, Airflow, MLflow, model serving): AI/ML engineers in Singapore earn S\$105K-130K, roughly 10-15% above equivalent data scientists, with MLOps/infrastructure depth cited as the main driver of the gap at senior level (skillup.sg AI/ML Engineer Salary Guide, 2026).
3. How to obtain it: SkillsFuture Credit + IMDA's TeSA programme subsidise this training for Singaporeans/PRs, lowering the cost of the AWS/GCP/Kubernetes route (skillup.sg, 2026).
4. Practical route: 1) one cloud associate certification (~2-3 months part-time), 2) a hands-on MLOps project for interviews, 3) target fintech/bank data-platform teams directly.

4. Financial feasibility check

Employees with 18-22 years' experience in this dataset average \$130K (range \$30K-\$219K, n=196,345). Narrowing to Finance/Web industry, senior-to-VP-level roles specifically, the average is \$141K - a +\$41K improvement over the current \$100K, comfortably clearing the \$102K annual-expense baseline. Even the more conservative FINANCE industry-wide average (\$129K) provides a \$27K surplus.

5. Leveraging personal interests

Board games and trading card games reflect analytical, probabilistic thinking - directly relevant to risk analytics and algorithmic trading in finance.